

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: C Batch: 3 Practical No: 2

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Subject Name : Deep Learning

Problem Statement : Design and implement a Multilayer Perceptron (MLP) for classification of the Iris or Wine dataset, and evaluate its performance using accuracy and a confusion matrix.

Algorithm :

1. Load the Wine dataset and separate features (X) and target labels (y).
2. Convert the feature data into a pandas DataFrame for inspection.
3. Print the first five rows to preview the data.
4. Plot histograms of all features to see their distributions.
5. Plot a scatter plot of alcohol vs. color intensity to inspect feature relationships.
6. Split the data into training and test sets (80/20).
7. Standardize the features using StandardScaler (fit on train, transform both).
8. Initialize an MLPClassifier with two hidden layers of 10 neurons each and ReLU activation.
9. Train the MLP on the scaled training data.
10. Predict class labels for the test set.
11. Compute the accuracy score by comparing predictions to true labels.
12. Compute the confusion matrix from predictions vs. true labels.
13. Print the accuracy, confusion matrix, and classification report (precision/recall/F1 per class).
14. Visualize the confusion matrix as an annotated heatmap.

Code:

```
# Import required libraries
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Wine dataset
wine = load_wine()
X = wine.data
y = wine.target

# Convert dataset into a DataFrame for visualization
df = pd.DataFrame(X, columns=wine.feature_names)

# Display first five rows of the dataset
print("First Five Rows of the Dataset:")
print(df.head())

# Histogram of all features
df.hist(figsize=(12, 8))
plt.tight_layout()
plt.show()

# Scatter plot of Alcohol vs Color Intensity
plt.figure(figsize=(6, 4))
plt.scatter(df["alcohol"], df["color_intensity"])
plt.xlabel("Alcohol")
plt.ylabel("Color Intensity")
plt.title("Alcohol vs Color Intensity")
plt.show()

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Normalize the data
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```

# Create the MLP model
mlp = MLPClassifier(
    hidden_layer_sizes=(10, 10),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)
# Train the model
mlp.fit(X_train, y_train)
# Make predictions
y_pred = mlp.predict(X_test)
# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("\nAccuracy:")
print(accuracy)

print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=wine.target_names
))

# Visualize the Confusion Matrix
plt.figure(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted Labels")
plt.ylabel("Actual Labels")
plt.title("Confusion Matrix")
plt.show()

```

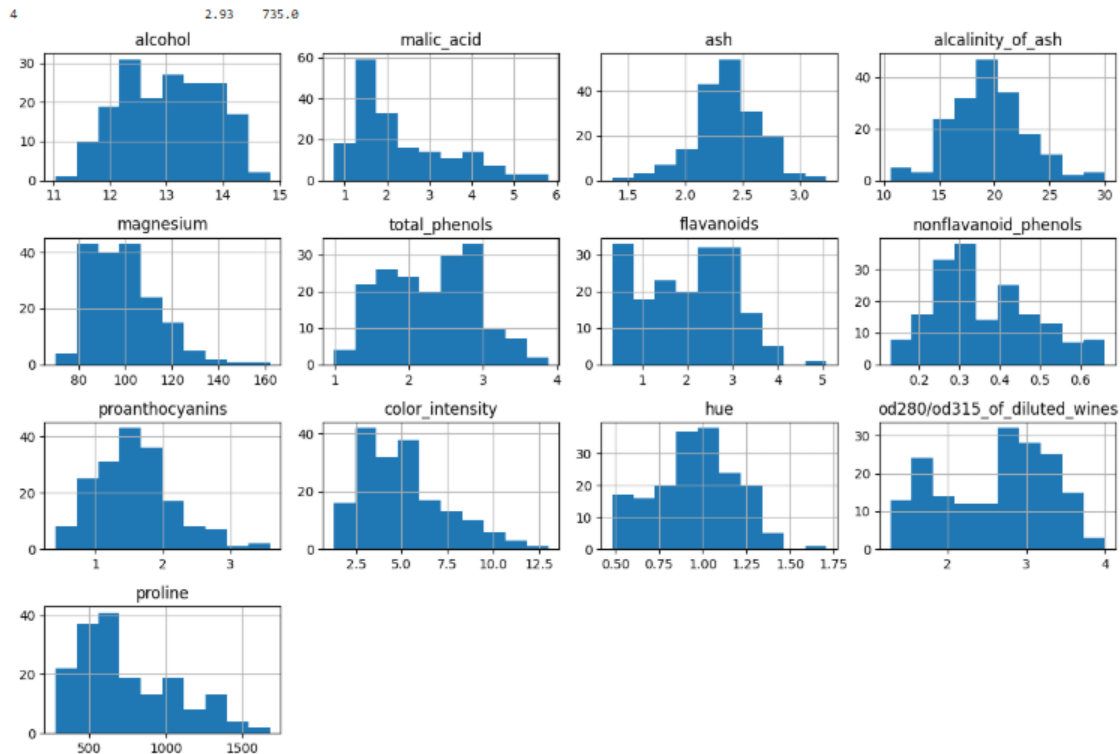
OUTPUT:

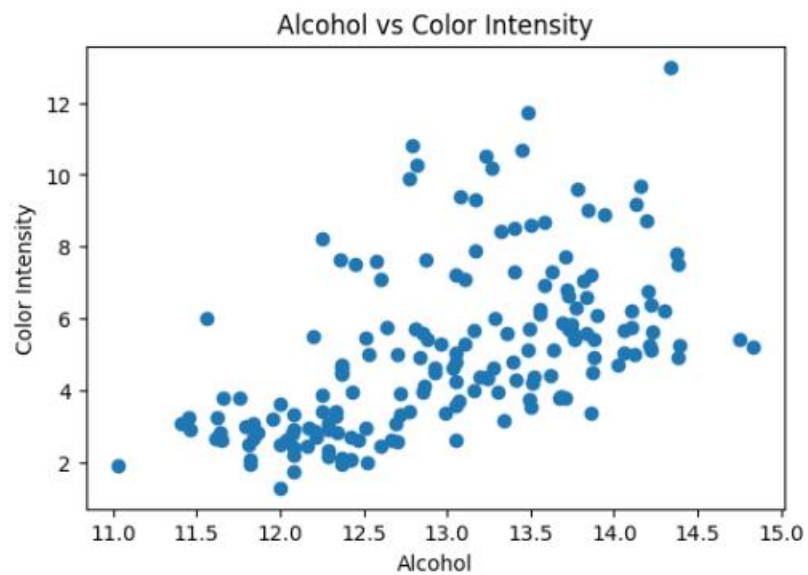
- First Five Rows of the Dataset:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	\
0	14.23	1.71	2.43	15.6	127.0	2.80	
1	13.20	1.78	2.14	11.2	100.0	2.65	
2	13.16	2.36	2.67	18.6	101.0	2.80	
3	14.37	1.95	2.50	16.8	113.0	3.85	
4	13.24	2.59	2.87	21.0	118.0	2.80	

	flavanoids	nonflavanoid_phenols	proanthocyanins	color_intensity	hue	\
0	3.06		0.28	2.29	5.64	1.04
1	2.76		0.26	1.28	4.38	1.05
2	3.24		0.30	2.81	5.68	1.03
3	3.49		0.24	2.18	7.80	0.86
4	2.69		0.39	1.82	4.32	1.04

	od280/od315_of_diluted_wines	proline
0	3.92	1065.0
1	3.40	1050.0
2	3.17	1185.0
3	3.45	1480.0
4	2.93	735.0





...

Accuracy:
1.0

Confusion Matrix:
[[14 0 0]
[0 14 0]
[0 0 8]]

Classification Report:

	precision	recall	f1-score	support
class_0	1.00	1.00	1.00	14
class_1	1.00	1.00	1.00	14
class_2	1.00	1.00	1.00	8
accuracy			1.00	36
macro avg	1.00	1.00	1.00	36
weighted avg	1.00	1.00	1.00	36

